

# COS20019-Cloud Computing Architecture



## Week 5: ACF Lab 5: RDS LAB

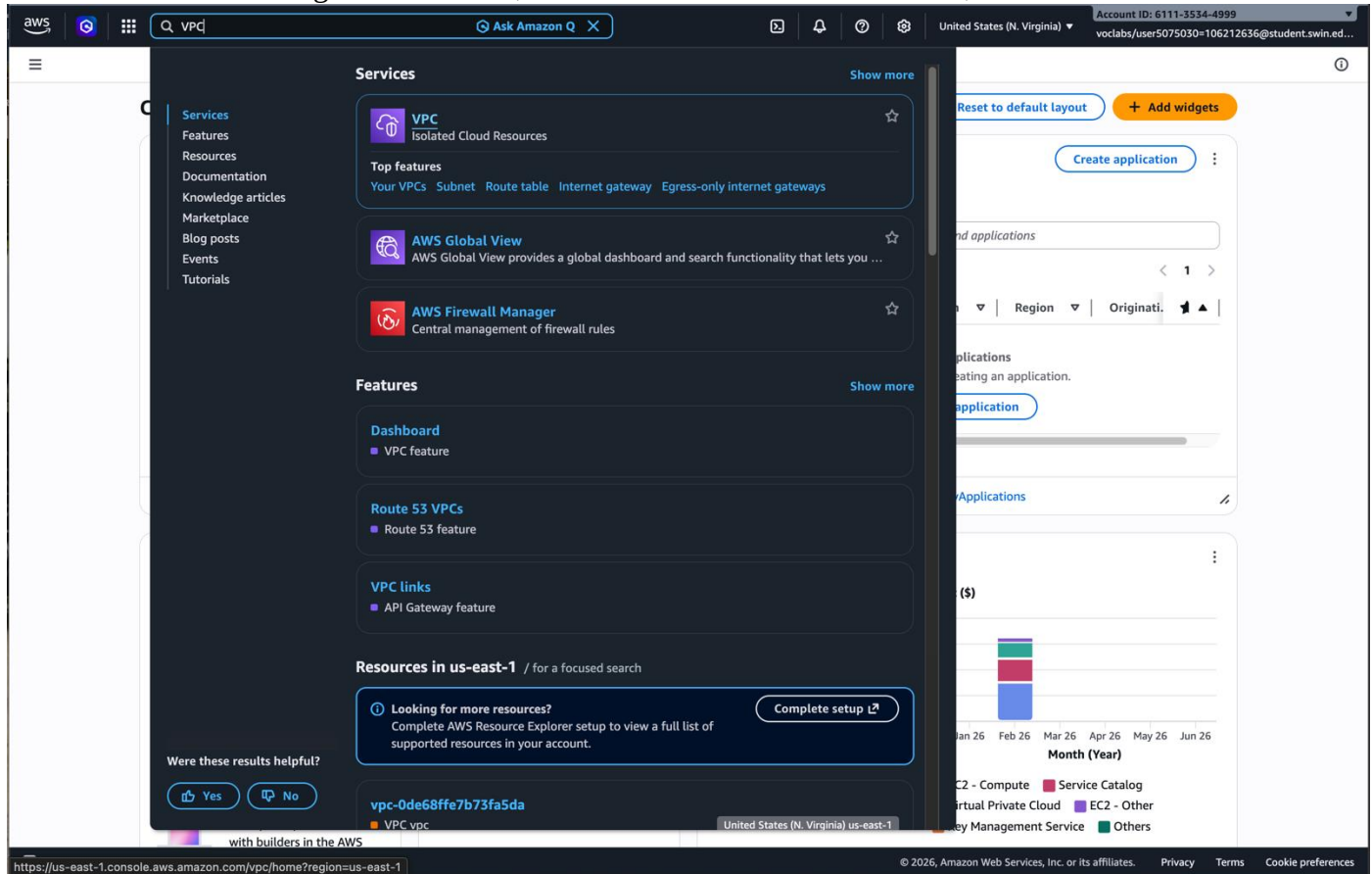
Author: Quang Huy Tran – Student ID: 106212636

Due date: 21/06/2026

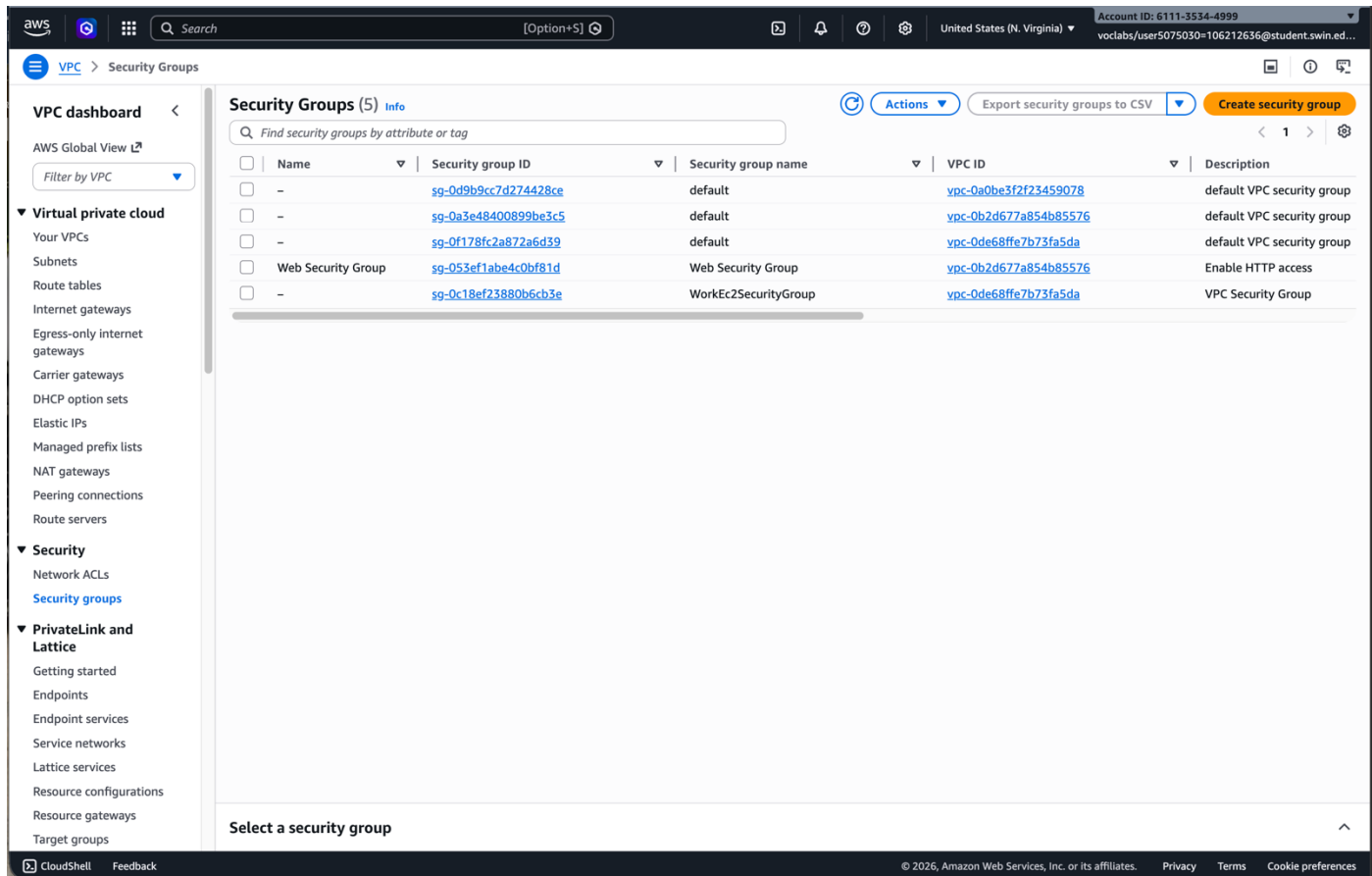
## Task 1: Create a Security Group for the RDS DB Instance

In this task, you will create a security group to allow your web server to access your RDS DB instance. The security group will be used when you launch the database instance.

3. In the AWS Management Console, in the search box next to **Services**, search for and select **VPC**.

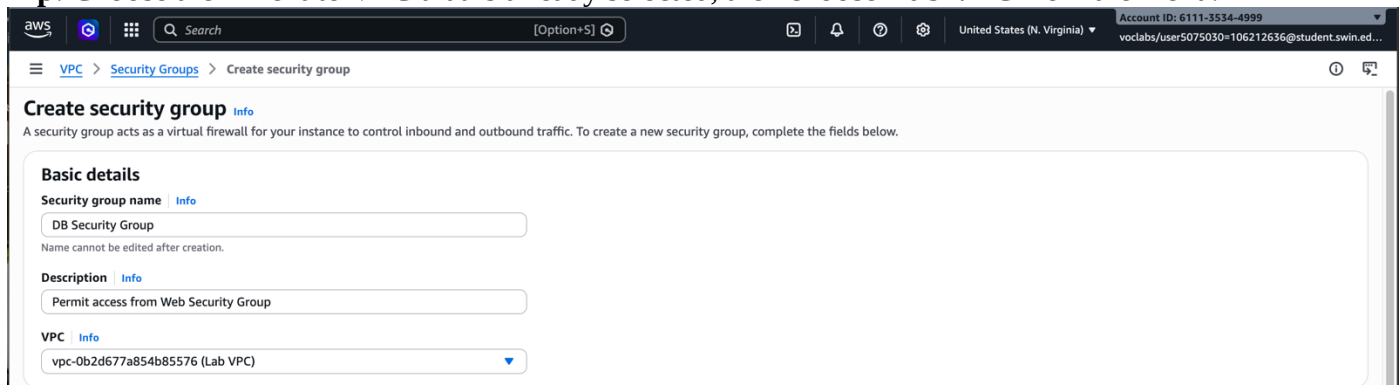


4. In the left navigation pane, choose **Security groups**.



5. Choose **Create security group** and then configure:
  - **Security group name:** DB Security Group
  - **Description:** Permit access from Web Security Group
  - **VPC:** Lab VPC

**Tip:** Choose the X next to VPC that is already selected, then choose **Lab VPC** from the menu.



6. In the **Inbound rules** pane, choose **Add rule**  
The security group currently has no rules. You will add a rule to permit access from the *Web Security Group*.

7. Configure the following settings:
  - **Type:** MySQL/Aurora (3306)
  - **Source:** Place your cursor in the field to the right of Custom, type sg, and then select *Web Security Group*.

This configures the Database security group to permit inbound traffic on port 3306 from any EC2 instance

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that is associated with the *Web Security Group*.

**Create security group** [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

**Basic details**

**Security group name** [Info](#)

DB Security Group

Name cannot be edited after creation.

**Description** [Info](#)

Permit access from Web Security Group

**VPC** [Info](#)

vpc-0b2d677a854b85576 (Lab VPC)

**Inbound rules** [Info](#)

| Type         | Protocol | Port range | Source               | Description - optional |
|--------------|----------|------------|----------------------|------------------------|
| MySQL/Aurora | TCP      | 3306       | Custom               |                        |
|              |          |            | sg-053ef1abe4c0bf81d |                        |
|              |          |            | sg-053ef1abe4c0bf81d |                        |

[Add rule](#) [Delete](#)

## 8. Choose **Create security group**

You will use this security group when launching an Amazon RDS database in this lab.

**Security group (sg-0287cd3f7d0a67d55 - DB Security Group) was created successfully**

**sg-0287cd3f7d0a67d55 - DB Security Group** [Actions](#)

**Details**

| Property             | Value                                 |
|----------------------|---------------------------------------|
| Security group name  | DB Security Group                     |
| Security group ID    | sg-0287cd3f7d0a67d55                  |
| Description          | Permit access from Web Security Group |
| VPC ID               | vpc-0b2d677a854b85576                 |
| Owner                | 611135344999                          |
| Inbound rules count  | 1 Permission entry                    |
| Outbound rules count | 1 Permission entry                    |

**Inbound rules (1)**

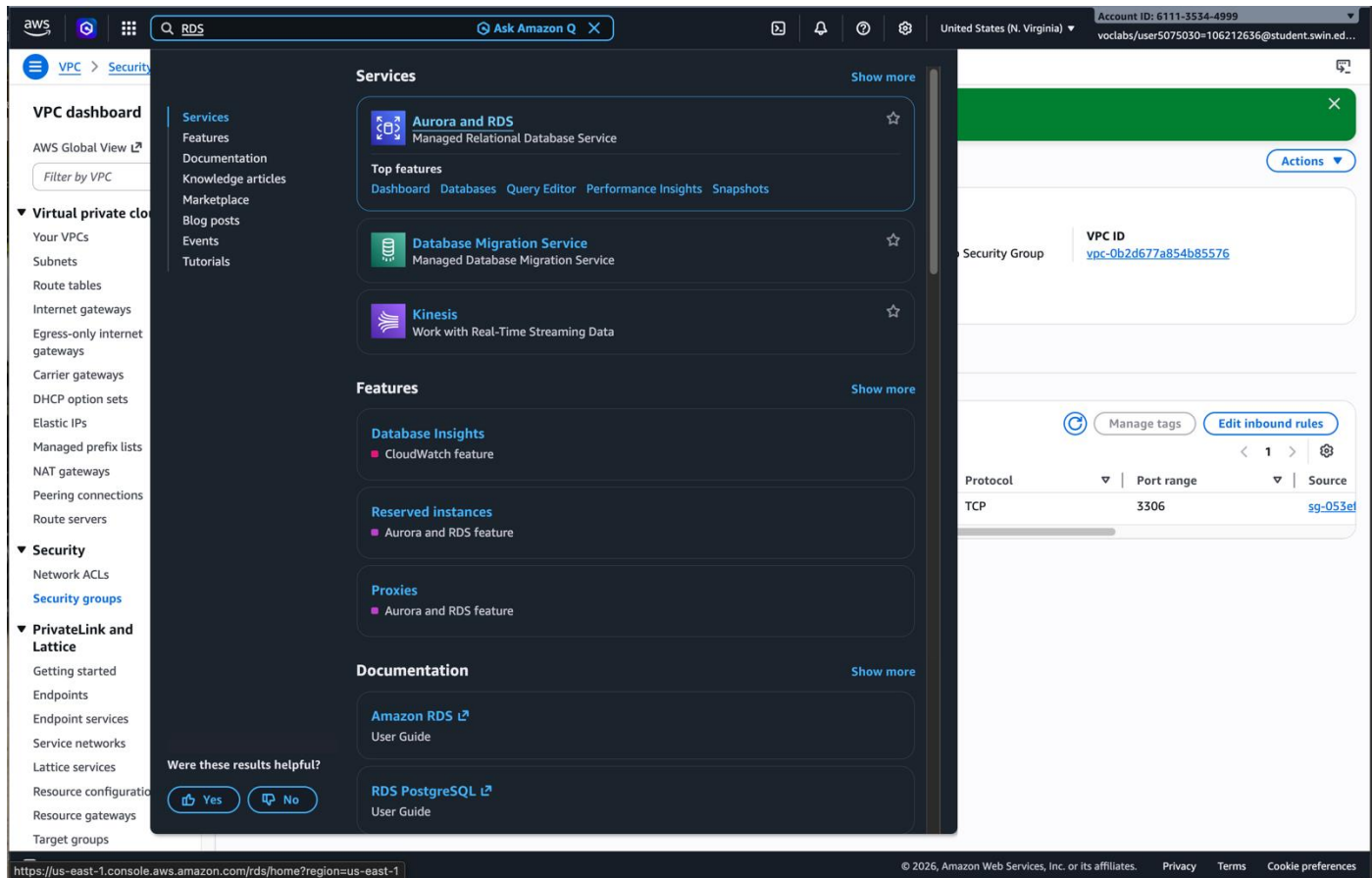
| Name | Security group rule ID | IP version | Type         | Protocol | Port range | Source               |
|------|------------------------|------------|--------------|----------|------------|----------------------|
| -    | sgr-06071a27e71b917c3  | -          | MySQL/Aurora | TCP      | 3306       | sg-053ef1abe4c0bf81d |

## Task 2: Create a DB Subnet Group

In this task, you will create a *DB subnet group* that is used to tell RDS which subnets can be used for the database. Each DB subnet group requires subnets in at least two Availability Zones.

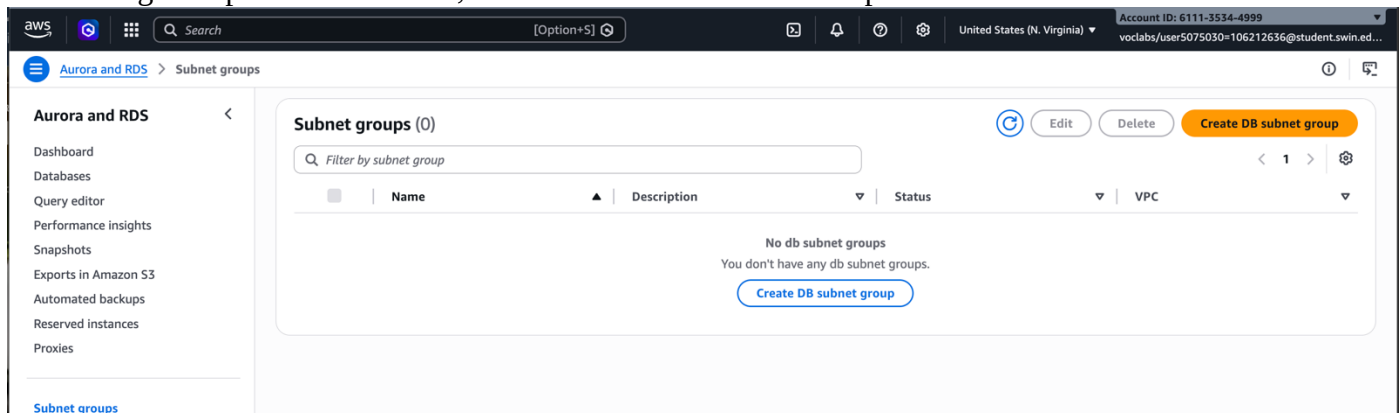
10. In the AWS Management Console, in the search box next to **Services**, search for and select **RDS**.

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11. In the left navigation pane, choose **Subnet groups**.

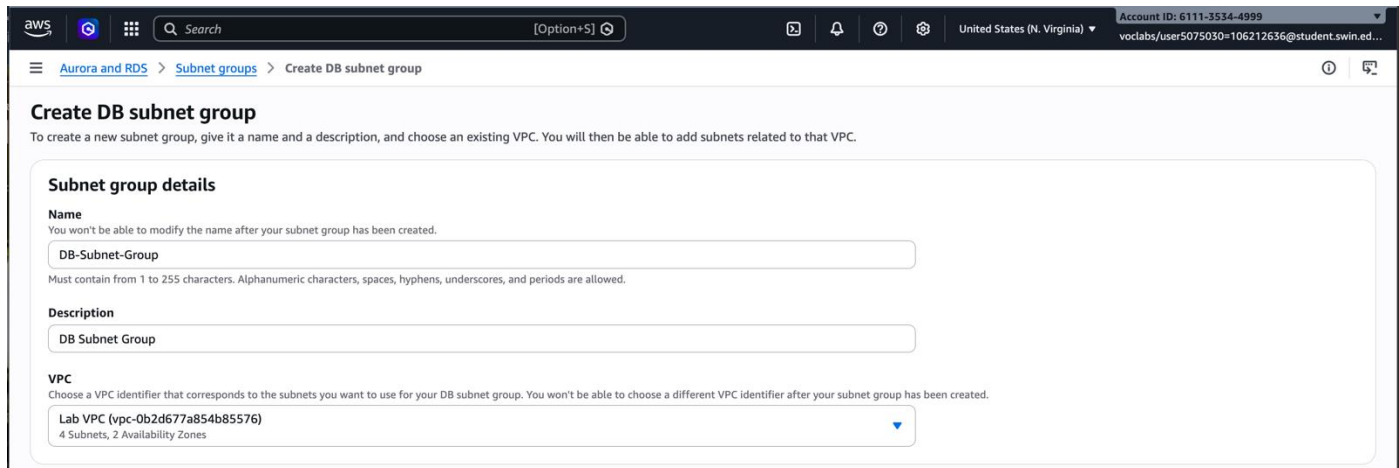
If the navigation pane is not visible, choose the menu icon in the top-left corner.



12. Choose **Create DB Subnet Group** then configure:

- **Name:** DB-Subnet-Group
- **Description:** DB Subnet Group
- **VPC:** Lab VPC

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**Create DB subnet group**

To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

**Subnet group details**

**Name**  
You won't be able to modify the name after your subnet group has been created.  
DB-Subnet-Group  
Must contain from 1 to 255 characters. Alphanumeric characters, spaces, hyphens, underscores, and periods are allowed.

**Description**  
DB Subnet Group

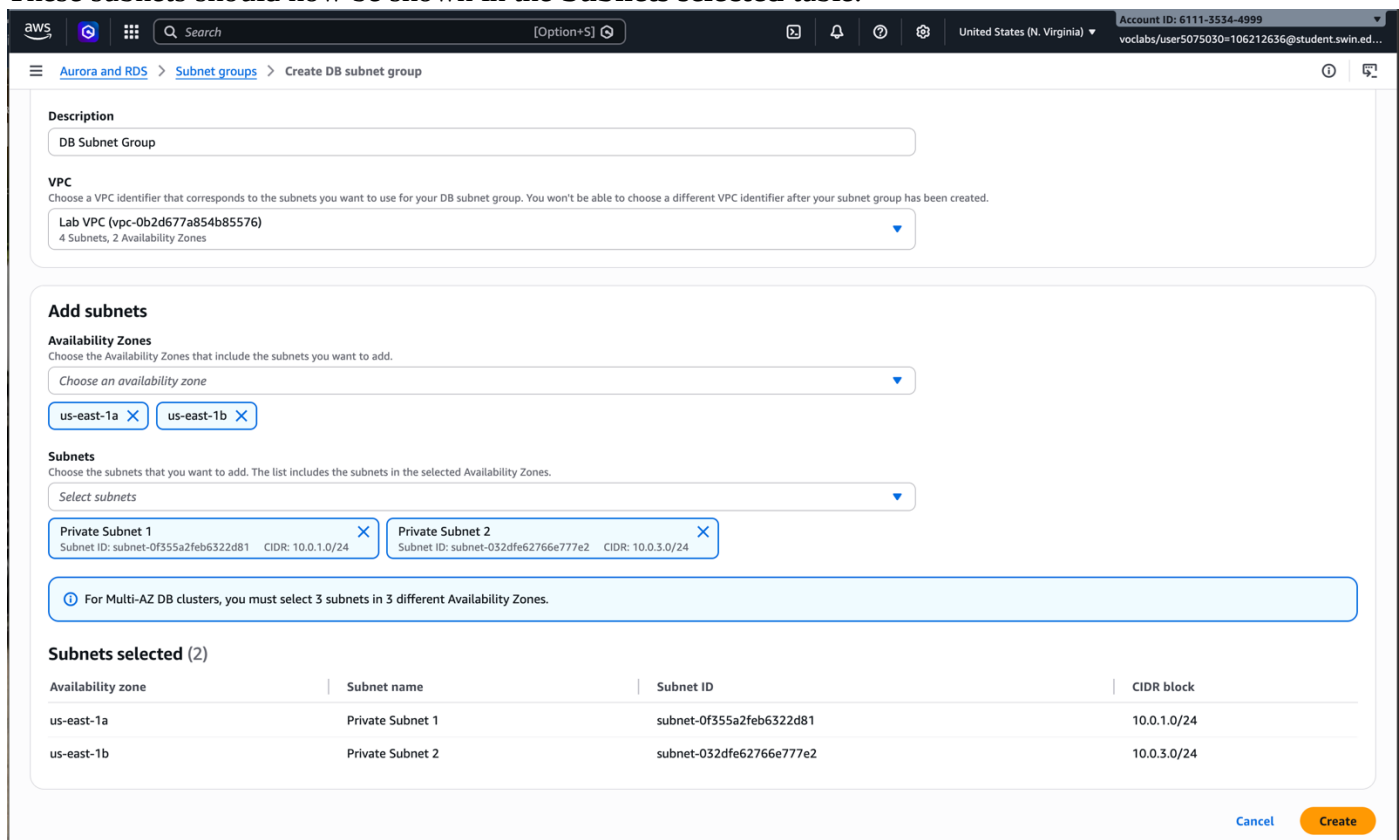
**VPC**  
Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.  
Lab VPC (vpc-0b2d677a854b85576)  
4 Subnets, 2 Availability Zones

13. Scroll down to the **Add subnets** section.

14. Expand the list of values under **Availability Zones** and select the first two zones: **us-east-1a** and **us-east-1b**.

15. Expand the list of values under **Subnets** and select the subnets associated with the CIDR ranges **10.0.1.0/24** and **10.0.3.0/24**.

These subnets should now be shown in the **Subnets selected** table.



**Add subnets**

**Availability Zones**  
Choose the Availability Zones that include the subnets you want to add.  
Choose an availability zone  
us-east-1a X us-east-1b X

**Subnets**  
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.  
Select subnets  
Private Subnet 1 X Private Subnet 2 X  
Subnet ID: subnet-0f355a2feb6322d81 CIDR: 10.0.1.0/24 Subnet ID: subnet-032dfe62766e777e2 CIDR: 10.0.3.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

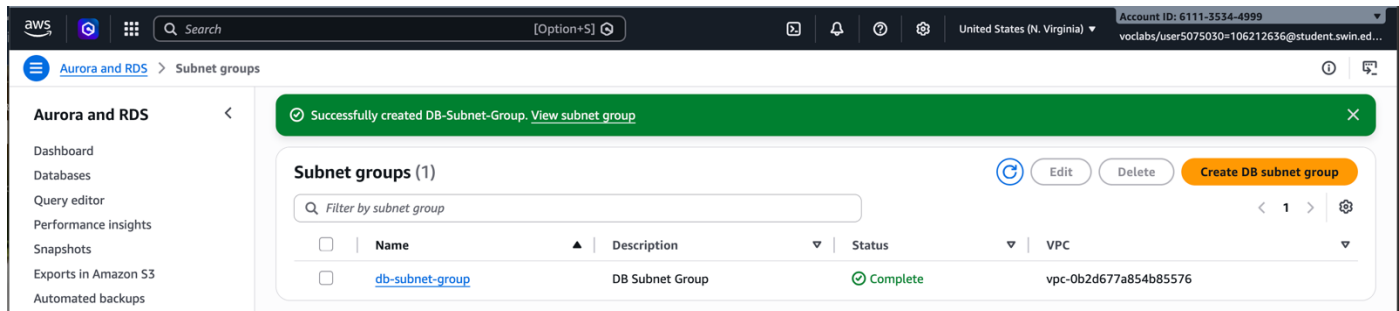
**Subnets selected (2)**

| Availability zone | Subnet name      | Subnet ID                | CIDR block  |
|-------------------|------------------|--------------------------|-------------|
| us-east-1a        | Private Subnet 1 | subnet-0f355a2feb6322d81 | 10.0.1.0/24 |
| us-east-1b        | Private Subnet 2 | subnet-032dfe62766e777e2 | 10.0.3.0/24 |

Cancel Create

16. Choose **Create**

You will use this DB subnet group when creating the database in the next task.

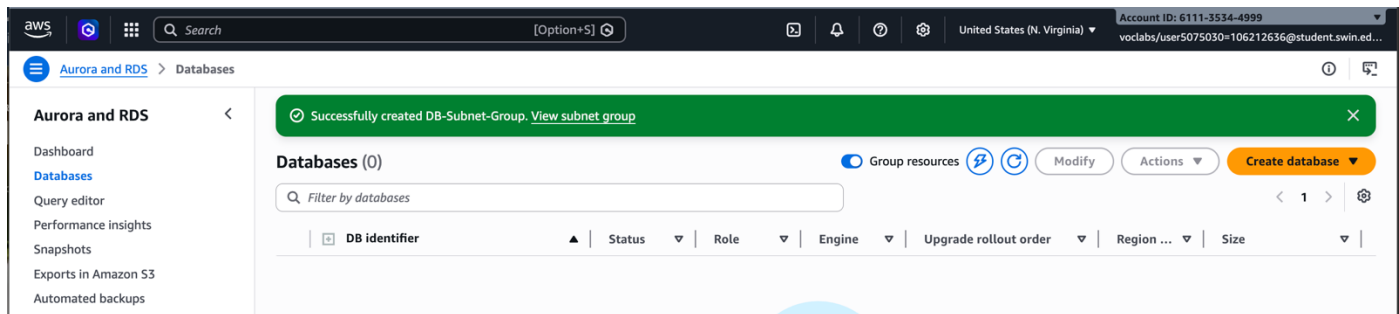


### Task 3: Create an Amazon RDS DB Instance

In this task, you will configure and launch a Multi-AZ Amazon RDS deployment of a MySQL database instance.

Amazon RDS **Multi-AZ** deployments provide enhanced availability and durability for Database (DB) instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

17. In the left navigation pane, choose **Databases**.



18. Choose **Create database**

If you see **Switch to the new database creation flow** at the top of the screen, please choose it.

19. Select **MySQL** under **Engine Options**.

20. Under **Templates** choose **Dev/Test**.

aws [Option+S] Ask Amazon Q United States (N. Virginia) Account ID: 6111-3534-4999 voclabs/user5075030=106212636@student.swin.ed...

Aurora and RDS > Databases > Create database

Successfully created DB-Subnet-Group. [View subnet group](#)

### Create database [Info](#)

#### Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 

☐ Aurora (PostgreSQL Compatible) 

☒ MySQL 

☐ PostgreSQL 

☐ MariaDB 

☐ Oracle 

☐ Microsoft SQL Server 

☐ IBM Db2 

#### Choose a database creation method

☒ Full configuration  
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create  
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

#### Templates

Choose a sample template to meet your use case.

☐ Production  
Use defaults for high availability and fast, consistent performance.

☒ Dev/Test  
This instance is intended for development use outside of a production environment.

☐ Free tier  
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

#### Availability and durability

Deployment options [Info](#)

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21. Under **Availability and durability** choose **Multi-AZ DB instance**.



aws [Option+S] Ask Amazon Q United States (N. Virginia) Account ID: 6111-3534-4999 voclabs/user5075030=106212636@student.swin.ed...

Aurora and RDS > Databases > Create database

### Availability and durability

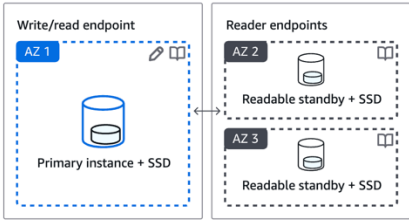
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

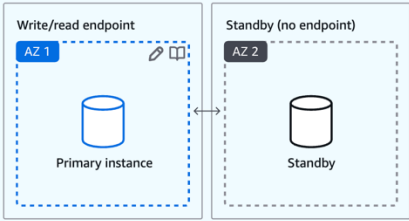
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



☒ Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

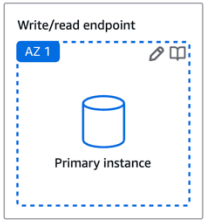
- 99.95% uptime
- Redundancy across Availability Zones



☐ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



### Settings

Edition

☒ MySQL Community

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)

Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☐ Show only versions that support the Amazon RDS Optimized Writes [Info](#)

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

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22. Under **Settings**, configure:

- **DB instance identifier:** lab-db
- **Master username:** main
- **Master password:** lab-password
- **Confirm password:** lab-password

**DB instance identifier** [Info](#)  
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.  
lab-db  
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

**Credentials Settings**

**Master username** [Info](#)  
Type a login ID for the master user of your DB instance.  
main  
1 to 16 alphanumeric characters. The first character must be a letter.

**Credentials management**  
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*  
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed  
Create your own password or have RDS create a password that you manage.

☐ Auto generate password  
Amazon RDS can generate a password for you, or you can specify your own password.

**Master password** [Info](#)  
\*\*\*\*\*

**Password strength** [Average](#)  
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' \* @

**Confirm master password** [Info](#)  
\*\*\*\*\*

**Additional credentials settings**

**Database authentication options** [Info](#)

☒ Password authentication  
Authenticates using database passwords.

☐ Password and IAM database authentication  
Authenticates using the database password and user credentials through AWS IAM users and roles.

☐ Password and Kerberos authentication  
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

23. Under **DB instance class**, configure:

- Select **Burstable classes (includes t classes)**.
- Select **db.t3.micro**

**Instance configuration**  
The DB instance configuration options below are limited to those supported by the engine that you selected above.

**DB instance class** [Info](#)

**Hide filters**

☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)  
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

**Instance type**

db.t3.micro  
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

24. Under **Storage**, configure:

- **Storage type:** *General Purpose (SSD)*
- **Allocated storage:** 20

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The screenshot shows the 'Storage' tab of the 'Create database' page in the AWS Management Console. The 'Storage type' is set to 'General Purpose SSD (gp3)' with a dropdown arrow. Below it, 'Allocated storage' is set to '20' GIB, with a range from 'Minimum: 20 GIB' to 'Maximum: 6,144 GIB'. 'Provisioned IOPS' is set to '3000'. A note states: 'Baseline IOPS of 3,000 IOPS is included for allocated storage less than 400 GIB.' The 'Storage throughput' section is partially visible at the bottom.

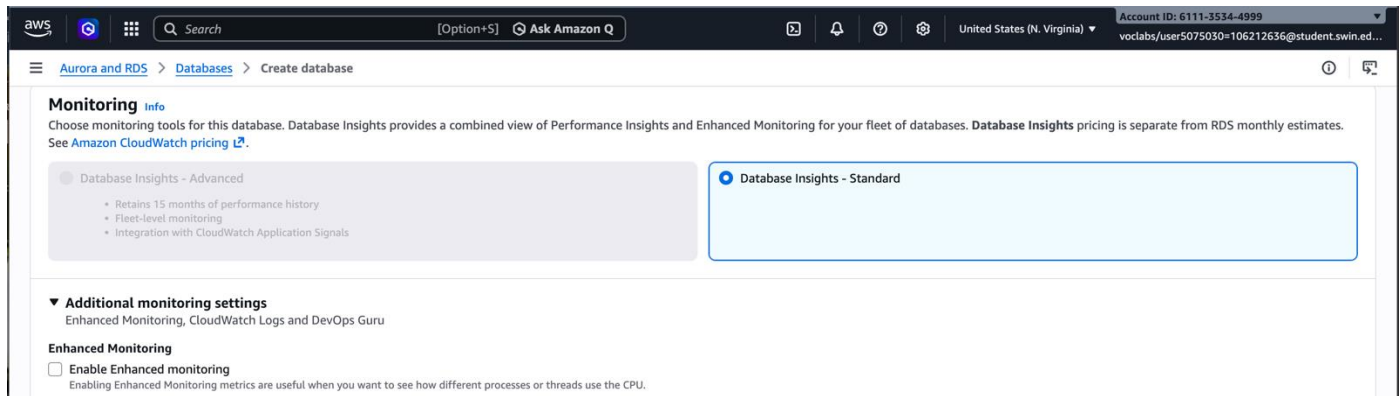
25. Under **Connectivity**, configure:
- **Virtual Private Cloud (VPC):** *Lab VPC*

The screenshot shows the 'Connectivity' tab of the 'Create database' page. Under 'Compute resource', the 'Don't connect to an EC2 compute resource' option is selected. Under 'Virtual private cloud (VPC)', the 'Lab VPC (vpc-0b2d677a854b85576)' is selected from a dropdown menu. A note at the bottom states: 'After a database is created, you can't change its VPC.'

26. Under **Existing VPC security groups**, from the dropdown list:
- Choose *DB Security Group*.
  - Deselect *default*

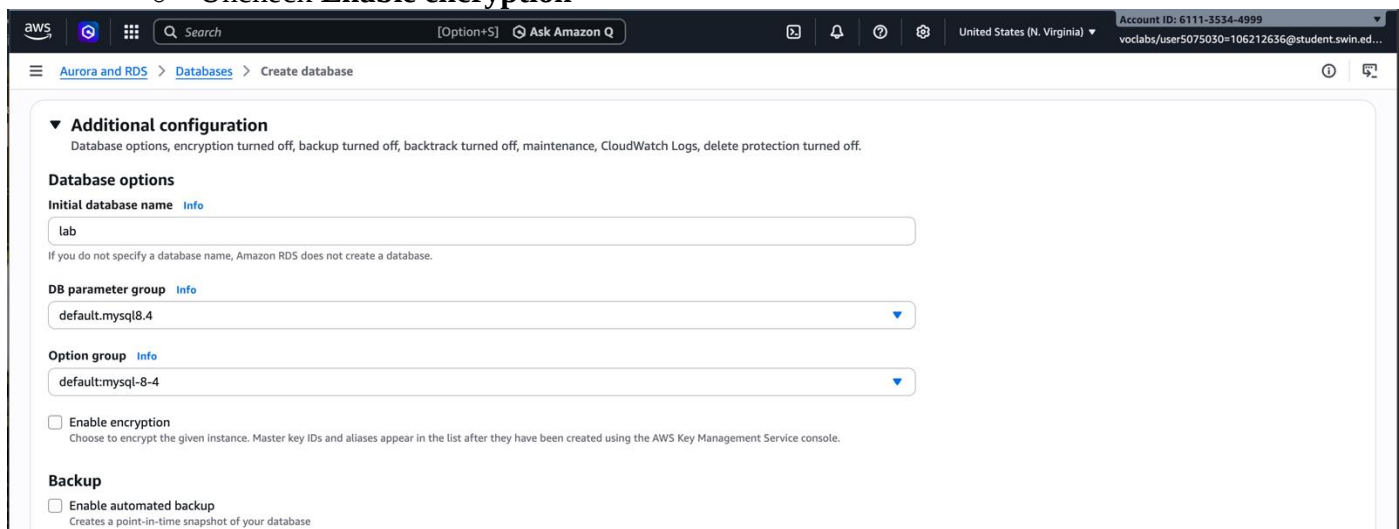
The screenshot shows the 'VPC security group (firewall)' tab. The 'Choose existing' option is selected. Under 'Existing VPC security groups', the dropdown menu is open, and 'DB Security Group' is selected, indicated by an 'X' icon next to the text.

27. Under **Monitoring** expand **Additional configuration**.
- Uncheck **Enable Enhanced monitoring**.



28. Under **Additional configuration**, configure:

- **Initial database name:** lab
- Uncheck **Enable automatic backups.**
- Uncheck **Enable encryption**

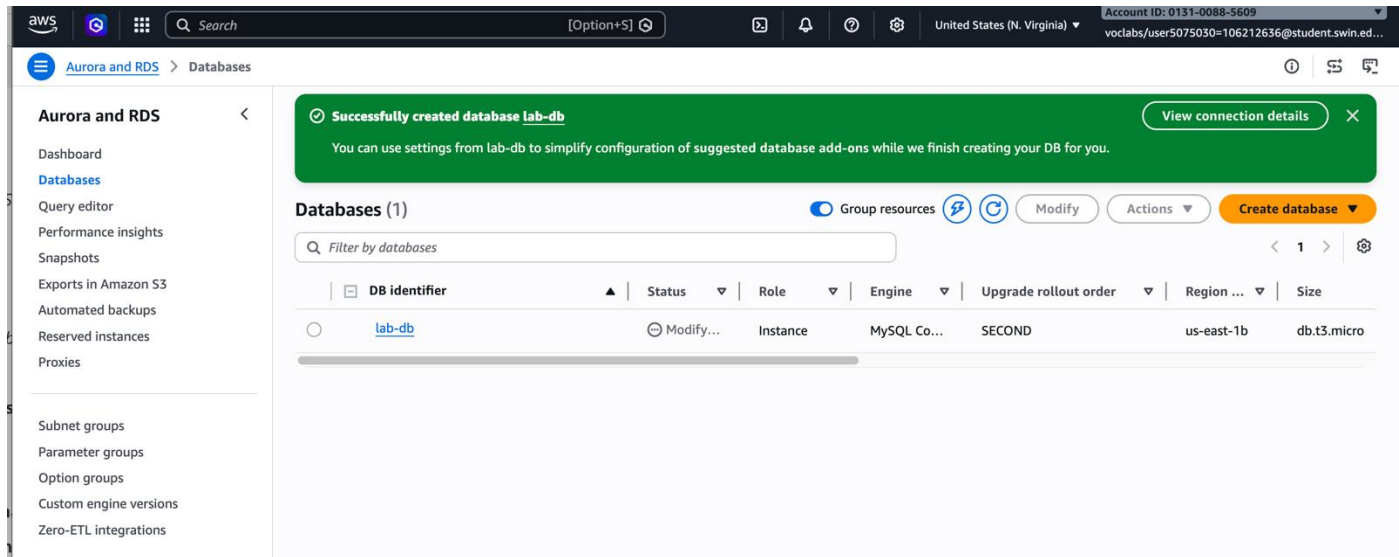


This will turn off backups, which is not normally recommended, but will make the database deploy faster for this lab.

29. Choose **Create database**

Your database will now be launched.

If you receive an error that mentions "not authorized to perform: iam:CreateRole", make sure you unchecked *Enable Enhanced monitoring* in the previous step.



30. Choose **lab-db** (choose the link itself).

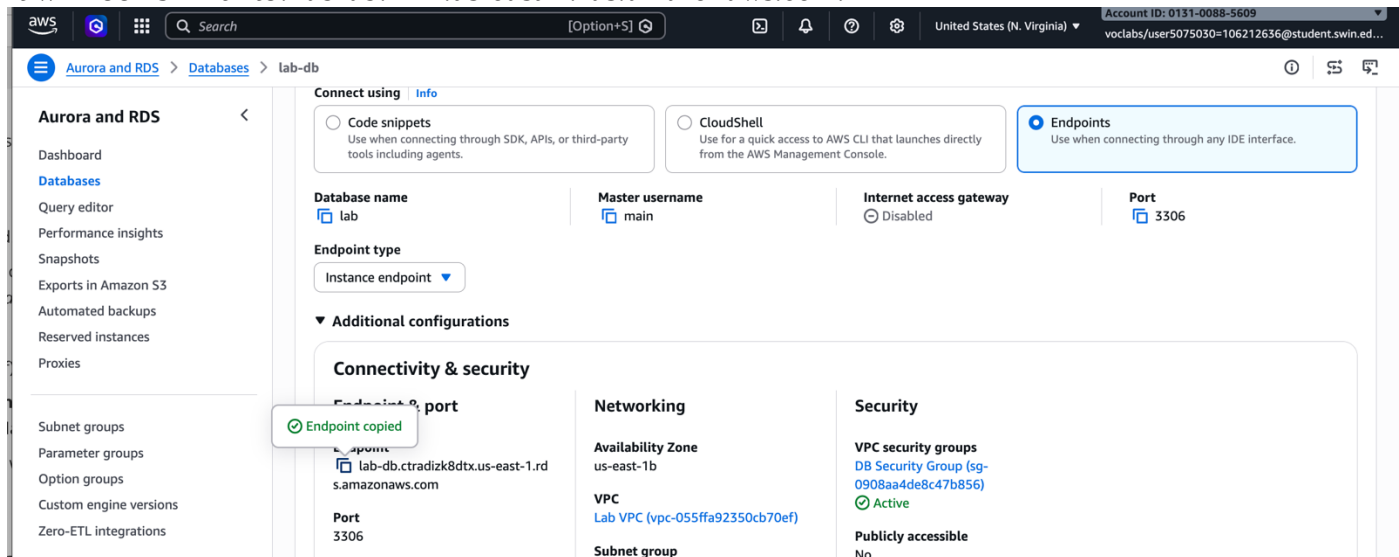
You will now need to wait **approximately 4 minutes** for the database to be available. The deployment process is deploying a database in two different Availability zones.

While you are waiting, you might want to review the [Amazon RDS FAQs](#) or grab a cup of coffee.

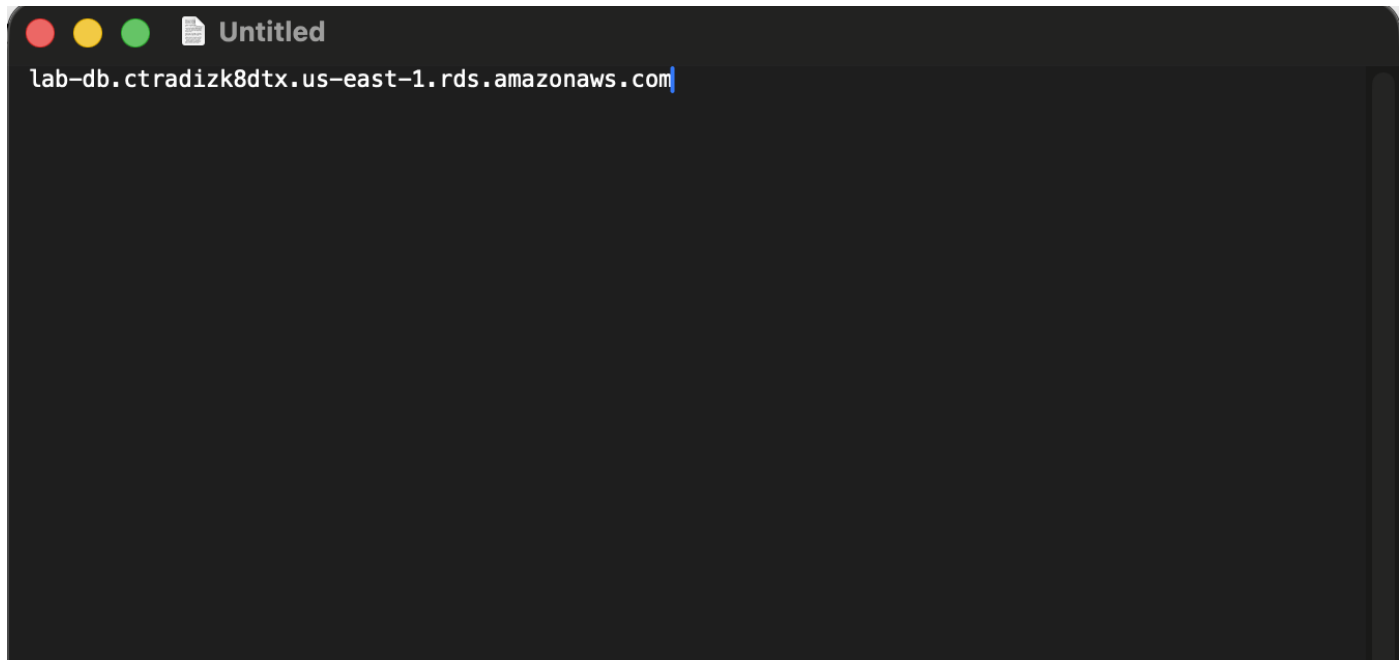
31. Wait until **Info** changes to **Modifying** or **Available**.

32. Scroll down to the **Connectivity & security** section and copy the **Endpoint** field.

It will look similar to: `lab-db.xxxx.us-east-1.rds.amazonaws.com`.



33. Paste the Endpoint value into a text editor. You will use it later in the lab.



#### Task 4: Interact with Your Database

In this task, you will open a web application running on a web server that has been created for you. You will configure it to use the database that you just created.

34. To discover the **WebServer** IP address, choose on the AWS Details drop down menu above these instructions. Copy the IP address value.

A screenshot of the AWS Cloud Labs interface. The top bar shows 'AWS' with a green dot, a timer at '01:12', and buttons for 'Start Lab', 'End Lab', 'AWS Details', and 'Details'. Below the top bar, there are buttons for 'Submit', 'Submission Report', and 'Grades'. The main content area is divided into two panels. The left panel shows step 29 of a lab, 'Choose Create database', with instructions for creating a database. The right panel shows 'Cloud Access' information, including the WebServer IP address 3.86.255.162. The interface is in English (EN-US).

29. Choose **Create database**

Your database will now be launched.

💬 If you receive an error that mentions "not authorized to perform: iam:CreateRole", make sure you unchecked *Enable Enhanced monitoring* in the previous step.

30. Choose **lab-db** (choose the link itself).

You will now need to wait **approximately 4 minutes** for the database to be available. The deployment process is deploying a database in two different Availability zones.

❗ While you are waiting, you might want to review the [Amazon RDS FAQs](#) or grab a cup of coffee.

31. Wait until **Info** changes to **Modifying** or **Available**.

32. Scroll down to the **Connectivity & security** section and copy the **Endpoint** field.

It will look similar to: `lab-db.xxxx.us-east-1.rds.amazonaws.com`.

**Cloud Access**

AWS CLI: **Show**

**Cloud Labs**

Remaining session time: 01:12:19(73 minutes)  
Session started at: 2026-06-12T03:44:40-0700  
Session to end at: 2026-06-12T05:14:40-0700

Accumulated lab time: 00:50:00 (50 minutes)

(1) ips -- public:3.86.255.162, private:10.0.2.38 (2) ips -- public:32.197.166.45, private:10.0.0.202

SSH key **Show** **Download PEM** **Download PPK**

AWS SSO **Download URL**

|             |                                         |
|-------------|-----------------------------------------|
| SecretKey   | BD451LZ5BXGEIQUxmyMsB9/gA4jsvrLmWHTdh5P |
| WebServer   | 3.86.255.162                            |
| BastionHost | 32.197.166.45                           |
| Region      | us-east-1                               |
| AccessKey   | AKIAQGDG7UZUSNZYJEFY                    |

35. Open a new web browser tab, paste the **WebServer** IP address and press Enter. The web application will be displayed, showing information about the EC2 instance.

---

 Load Test RDS

| Meta-Data         | Value               |
|-------------------|---------------------|
| InstanceId        | i-0ea3c6fb8da702c1b |
| Availability Zone | us-east-1b          |

Current CPU Load: **0%**

36. Choose the **RDS** link at the top of the page.  
You will now configure the application to connect to your database.

---

 Load Test RDS

Endpoint

Database

Username

Password

Submit

37. Configure the following settings:

- **Endpoint:** Paste the Endpoint you copied to a text editor earlier
- **Database:** lab
- **Username:** main
- **Password:** lab-password
- Choose **Submit**

A message will appear explaining that the application is running a command to copy information to the database. After a few seconds the application will display an **Address Book**.  
The Address Book application is using the RDS database to store information.

Endpoint: lab-db.ctradizk8dtx.us-east-1.rds.amazonaws.com

Database: lab

Username: main

Password: .....

Submit

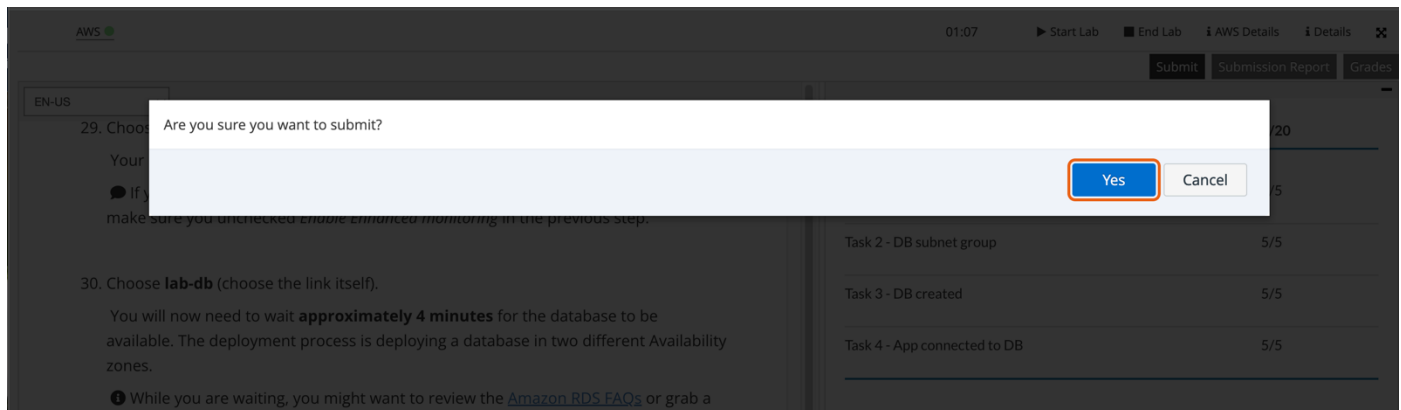
38. Test the web application by adding, editing and removing contacts.  
The data is being persisted to the database and is automatically replicating to the second Availability Zone.

### Address Book

| Last name | First name | Phone        | Email                      | Admin                                       |
|-----------|------------|--------------|----------------------------|---------------------------------------------|
| Doe       | Jane       | 010-110-1101 | janed@someotheraddress.org | <a href="#">Add Contact</a>                 |
| Johnson   | Roberto    | 123-456-7890 | robertoj@someaddress.com   | <a href="#">Edit</a> <a href="#">Remove</a> |

## Submitting your work

39. To record your progress, choose **Submit** at the top of these instructions.



40. When prompted, choose **Yes**.

After a couple of minutes, the grades panel appears and shows you how many points you earned for each task. If the results don't display after a couple of minutes, choose **Grades** at the top of these instructions.  
**Tip:** You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is recorded for this lab.



The screenshot shows the AWS Lab interface. On the left, step 29 is titled 'Create database' and includes instructions for launching a database and waiting for it to be available. On the right, a table displays the completion status of four tasks.

| Task                            | Score |
|---------------------------------|-------|
| Total score                     | 20/20 |
| Task 1 - Security Group created | 5/5   |
| Task 2 - DB subnet group        | 5/5   |
| Task 3 - DB created             | 5/5   |
| Task 4 - App connected to DB    | 5/5   |

41. To find detailed feedback about your work, choose **Submission Report**.

**Tip:** For any checks where you did not receive full points, there are sometimes helpful details provided in the submission report.

## Lab Complete

Congratulations! You have completed the lab.

42. Choose End Lab at the top of this page and then choose **Yes** to confirm that you want to end the lab. A panel will appear, indicating that "DELETE has been initiated... You may close this message box now."

The screenshot shows the AWS Lab interface with a confirmation dialog box overlaid. The dialog asks 'Are you sure you want to end the lab?' and includes a warning that resources will be permanently deleted. The 'Yes' button is highlighted with a red box.

43. Choose the **X** in the top right corner to close the panel.